

Module4_Lab2

September 25, 2026

```
[1]: import pandas as pd
df = pd.read_csv("allbacks.csv")
df.head()
```

```
[1]:  rownames  volume  area  weight  cover
0         1     885   382     800     hb
1         2    1016   468     950     hb
2         3    1125   387    1050     hb
3         4     239   371     350     hb
4         5     701   371     750     hb
```

```
[2]: df.shape()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[2], line 1
----> 1 df.shape()

TypeError: 'tuple' object is not callable
```

```
[3]: df.shape
```

```
[3]: (15, 5)
```

```
[4]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   rownames    15 non-null     int64
1   volume      15 non-null     int64
2   area        15 non-null     int64
3   weight      15 non-null     int64
4   cover       15 non-null     str
dtypes: int64(4), str(1)
memory usage: 762.0 bytes
```

```
[5]: df.describe()
```

```
[5]:
```

	rownames	volume	area	weight
count	15.000000	15.000000	15.000000	15.000000
mean	8.000000	845.266667	182.800000	706.666667
std	4.472136	339.879909	203.586837	268.837887
min	1.000000	239.000000	0.000000	250.000000
25%	4.500000	618.000000	0.000000	512.500000
50%	8.000000	929.000000	0.000000	725.000000
75%	11.500000	1025.000000	376.500000	950.000000
max	15.000000	1492.000000	468.000000	1075.000000

```
[6]: data = [hb, hb, hb, hb, hb, hb, hb, pb, pb, pb, pb, pb, pb, pb, pb]
median = statistics.median(data)
print(median)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[6], line 1
----> 1 data = [hb, hb, hb, hb, hb, hb, hb, pb, pb, pb, pb, pb, pb, pb, pb]
      2 median = statistics.median(data)
      3 print(median)

NameError: name 'hb' is not defined
```

```
[7]: data = ["hb", "hb", "hb", "hb", "hb", "hb", "hb", "pb", "pb", "pb", "pb", "pb", "pb", "pb", "pb"]
      ↪ "pb", "pb", "pb"]
median = statistics.median(data)
print(median)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[7], line 2
      1 data = ["hb", "hb", "hb", "hb", "hb", "hb", "hb", "pb", "pb", "pb", "pb", "pb", "pb", "pb", "pb", "pb"]
      ↪ "pb", "pb", "pb", "pb", "pb"]
----> 2 median = statistics.median(data)
      3 print(median)

NameError: name 'statistics' is not defined
```

```
[8]: import statistics
data = ["hb", "hb", "hb", "hb", "hb", "hb", "hb", "pb", "pb", "pb", "pb", "pb", "pb", "pb", "pb", "pb"]
      ↪ "pb", "pb", "pb"]
median = statistics.median(data)
print(median)
```

```
Cell In[8], line 1
    import as statistics
    ~
SyntaxError: invalid syntax
```

```
[9]: import statistics
data = ["hb", "hb", "hb", "hb", "hb", "hb", "hb", "pb", "pb", "pb", "pb", "pb", "pb", "pb", "pb", "pb"]
median = statistics.median(data)
print(median)
```

pb

Since the Cover data set have odd numbers, the median is the middle point, which is the number 8 item on the list. The median is pb.

```
[10]: mean = df["weight"].mean()
print(mean)
```

706.6666666666666

Total mean weight of all the books is 706.67

```
[11]: hb_data = [800, 950, 1050, 350, 750, 600, 1075]
mean = sum(hb_data) / len(hb_data)
print(mean)
```

796.4285714285714

```
[12]: pb_data = [250, 700, 650, 975, 350, 950, 425, 725]
mean = sum(pb_data) / len(pb_data)
print(mean))
```

```
Cell In[12], line 3
    print(mean))
    ~
SyntaxError: unmatched ')'
```

```
[13]: pb_data = [250, 700, 650, 975, 350, 950, 425, 725]
mean = sum(pb_data) / len(pb_data)
print(mean)
```

628.125

Mean weight of hb is 796.4 and mean weight of pb is 628.1

```
[14]: variance = df["volume"].var()  
      print(variance)
```

```
115518.3523809524
```

The variance in book volumes is 115,518

Decide which book cover types the library should opt for in the future

Weight-wise, obviously the paperback books are lighter. Might be better for those going to the library to carry the books around versus hardcover books, which are heavier in weight. Paperback books also tend to be cheaper.

```
[ ]:
```